The Divisibility Obstruction to a Game-Native Clifford Deformation

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Abstract

The exterior algebra of the short-game group admits many quadratic deformations after one first chooses a small subgroup and then supplies a quadratic table by hand. That local freedom does not extend to a quadratic datum belonging to game values themselves. The reason is the abstract group structure of short games: it is 2-divisible and all of its torsion is 2-primary. We prove that in any coefficient-faithful Clifford realization of such a group, every torsion game has square zero and is polar-orthogonal to every game. Equivalently, every globally defined, or subgroup-inclusion coherent, quadratic datum factors through the torsion-free quotient. This gives a negative answer under an exact naturality condition and explains why the nonzero Gold forms on nimbers cannot extend to general partizan games.

1 Introduction and main result

The implemented object `GameExterior` starts with finitely many short games, takes their $\mathbb{Z}$-span M, and forms $\Lambda_{\mathbb{Z}} M$. The checked `GameClifford` surface accepts integer values

$$e_x^2 = Q(x), \qquad e_x e_y + e_y e_x = B(x, y)$$

only after every known relation in M is null and polar-radical. Its tables are intentionally supplied by the caller. This leaves a mathematical question: can the game structure itself provide a nonzero torsion square, rather than merely allowing a table on one selected presentation?

The phrase "game-native" needs a naturality condition. Without one, the question has the tautological answer "yes": on $\langle * \rangle \cong \mathbb{Z}/2$ one may choose characteristic-two coefficients and declare $e_*^2 = 1$. Nothing in that declaration distinguishes $*$ from an abstract generator of $\mathbb{Z}/2$, and the value changes when the ambient subgroup changes.

Definition 1 (ambient coherence)**.** A Clifford deformation of short games is *ambient-coherent* if it is either a single coefficient-faithful datum on the full additive group ShUg, or, sufficiently for finite calculations, a directed family of data on finitely generated subgroups $M \leq \mathrm{ShUg}$ such that every inclusion $M \subseteq N$ induces injective coefficient and algebra maps preserving the grade-one generators, their squares, and their polar anticommutators.

This is the weakest contract under which $Q(x)$ is attached to the game value x rather than to the accident that one stopped at a root-incomplete subgroup. It permits the coefficient ring and algebra to grow with the subgroup. Injectivity is essential: without it an inclusion may "preserve" a square only by killing the coefficient that recorded it.

2 Divisibility of the short-game group

Let $\mathbb{Z}[1/2] = \mathbb{Z}[1/2]$ be the additive group of dyadic rationals.

Theorem 2 (Moews). *The additive group of short partizan games has abstract group structure*

$$\mathrm{ShUg} \cong \bigoplus_{\mathbb{N}} \mathbb{Z}[1/2] \;\oplus\; \bigoplus_{\mathbb{N}} \mathbb{Z}[1/2]/\mathbb{Z}.$$

Moreover, every finite-order short game has order a power of two.

The order statement is Theorem 8 and the decomposition is Theorem 9 of Moews [1]. Two consequences, and only these consequences, enter the new proof:

(i) multiplication by 2^k on ShUg is surjective for every k;

(ii) if t is torsion, then $2^k t = 0$ for some k.

Notice that ShUg is not divisible by every integer: the $\mathbb{Z}[1/2]$ summands are not 3-divisible. Full divisibility is neither claimed nor needed.

3 The algebraic root lemma

The obstruction occurs before any classification of quadratic maps. It is a statement about an additive grade-one realization in an arbitrary associative ring.

Lemma 3 (root collapse). *Let A be an abelian group, C an associative ring, and $i : A \to (C, +)$ an additive map. Suppose $nt = 0$ and choose $y, z \in A$ with*

$$ny = t, \qquad nz = x.$$

Then

$$i(t)^2 = 0, \qquad i(t)i(x) + i(x)i(t) = 0.$$

Proof. The relation $nt = 0$ and the identity $ny = t$ give

$$n^2 y = n(ny) = nt = 0,$$

hence $n^2 i(y) = i(n^2 y) = 0$. Since integer scalars are central in C,

$$\begin{aligned} i(t)^2 &= (ni(y))(ni(y)) = n^2 i(y)^2 = (n^2 i(y))i(y) = 0, \\ i(t)i(x) + i(x)i(t) &= (ni(y))(ni(z)) + (ni(z))(ni(y)) \\ &= n^2\big(i(y)i(z) + i(z)i(y)\big) \\ &= (n^2 i(y))i(z) + i(z)(n^2 i(y)) = 0. \end{aligned}$$

No commutativity of C, quadratic homogeneity axiom, or assumption on its characteristic is used. □

Remark 4 (formal proof). The kernel-checked file is `GameExterior.lean`. Its theorem takes the roots y, z explicitly, proves both ring identities, and then proves the coefficient-valued corollaries below from injectivity of the algebra map. The file contains no `sorry`, custom axiom, or finite computation. The Moews theorem remains an explicit external input rather than being encoded as an axiom.

4 Global torsion-radical collapse

Let R be a commutative ring and C an R-algebra. A coefficient-faithful Clifford datum consists of an additive grade-one map $i : A \to C$, functions $Q : A \to R$ and $B : A \times A \to R$, relations

$$i(x)^2 = \iota(Q(x)), \qquad i(x)i(y) + i(y)i(x) = \iota(B(x, y)),$$

and an injective coefficient map $\iota : R \hookrightarrow C$.

Theorem 5 (game-torsion radical theorem)**.** *In every ambient-coherent coefficient-faithful Clifford datum on short games,*

$$Q(t) = 0, \qquad B(t, x) = B(x, t) = 0$$

for every torsion game t and every short game x. Consequently

$$Q(x + t) = Q(x)$$

and the quadratic datum factors uniquely through $\mathrm{ShUg}\,/\,\mathrm{Tor}(\mathrm{ShUg})$.

Proof. First take a global datum. By Theorem 2, choose a power of two n such that $nt = 0$, and choose n-th roots $ny = t$ and $nz = x$. Apply Lemma 3. The square and anticommutator relations give

$$\iota(Q(t)) = 0, \qquad \iota(B(t, x)) = 0.$$

Injectivity of ι gives the two asserted vanishings. Symmetry of the displayed anticommutator gives $B(x, t) = 0$ as well. Finally, the defining polarization identity gives

$$Q(x + t) = Q(x) + Q(t) + B(x, t) = Q(x).$$

For a directed family, begin in any finitely generated subgroup containing t and x. Choose the roots y, z in ShUg and enlarge to the finitely generated subgroup containing t, x, y, z. The preceding proof applies there. The injective transition maps pull the vanishing back to the original subgroup. This is exactly where ambient coherence is used. □

Corollary 6 (coefficient-independent no-go)**.** *Changing the commutative coefficient target cannot rescue a game-native torsion square. The conclusion of Theorem 5 holds in characteristic zero, characteristic two, and torsion rings such as $\mathbb{Z}/4$.*

Proof. The proof of Lemma 3 is an identity in the target algebra and does not inspect the characteristic of R. □

5 Consequences for the proposed escapes

5.1 The characteristic-two local square

On the isolated subgroup $M = \langle * \rangle \cong \mathbb{Z}/2$, the declaration $e_*^2 = 1$ over $\mathbb{F}_2$ is a valid local Clifford algebra. Moews's theorem supplies a short partizan game h with $2h = *$. In any coherent enlargement containing h,

$$e_*^2 = (2e_h)^2 = 4e_h^2 = 0.$$

Thus the local square cannot survive an injective transition map. It is a property of the root-incomplete presentation $\langle * \rangle$, not a quadratic invariant of the game value $*$.

The same argument eliminates the universal local record $\mathrm{Sym}_{\mathbb{F}_2}(M/2M)$. Globally, $\mathrm{ShUg} = 2\,\mathrm{ShUg}$, so $\mathrm{ShUg}\,/2\,\mathrm{ShUg} = 0$.

5.2 The Brown $\mathbb{Z}/4$ route

The older coefficient-faithfulness obstruction already shows that an odd local value $Q(*) \in \mathbb{Z}/4$ collapses the scalar 2. Theorem 5 is stronger: even a locally faithful even value cannot extend coherently, because the partizan roots of $*$ force $Q(*) = 0$ in every coefficient ring. Hence a genuine odd Brown phase cannot arise as the square of a globally game-native Clifford generator. Brown quadratic modules remain valid objects on $\mathbb{F}_2$-spaces; this theorem concerns their proposed extension as faithful Clifford square data on all short games.

5.3 Why the Gold forms stop at the nimber core

Finite nimber fields carry extra multiplication and trace, producing the nonzero forms

$$Q_a(u) = \mathrm{Tr}(u\, u^{2^a}).$$

These are genuinely non-tautological on the impartial field-like core. Every additive nimber is nevertheless a torsion short game. Therefore Theorem 5 implies that no nonzero Gold form extends to an ambient-coherent coefficient-faithful Clifford datum on general short games. The obstruction is not the absence of a clever coefficient ring; adjoining the partizan halves of nimbers kills the torsion quadratic data.

5.4 What survives

Quadratic data on the torsion-free quotient survives. The implemented mean-value and atomic-weight constructions are examples of additive game invariants whose squares live on free directions. The theorem predicts their documented behavior: they are blind to torsion.

6 Conclusion

The commutative-scalar Clifford architecture has a negative answer. A quadratic datum belonging to short-game values must be ambient-coherent; every such coefficient-faithful datum is Grassmann on the entire torsion subgroup and factors through the torsion-free quotient. This is a complete obstruction, not a finite search or a restriction to integer coefficients.

The theorem deliberately does not forbid three different objects:

1. a hand-supplied table on one chosen root-incomplete subgroup;
2. the nonzero Gold forms retained inside the nimber field-like core; or
3. a directed or noncommutative game construction that is not a Clifford algebra over a commutative coefficient ring with additive grade one.

The first is exactly the existing engineering API and is not game-native under Definition 1; the second cannot extend beyond its core; the third changes the mathematical object and belongs to directed game semantics, not to a deformation of `GameExterior`. No commutative coefficient target evades the theorem.

References

[1] David Moews, *The Abstract Structure of the Group of Games*, in *More Games of No Chance*, MSRI Publications 42 (2002), 49–58. `https://library.slmath.org/books/Book42/files/moews.pdf`.